Son of Surya

GUIDE
SOS-01
FUNDAMENTALS

FREE STARTER GUIDE | BEGINNER

# Solar Starter Guide

Understand solar before you buy it — system types, components, the Indian rooftop journey, subsidies at a glance, first size and savings, and the questions that protect you.

| On | Hybrid | Off | 8 | ₹ |
|---|---|---|---|---|
| GRID | BACKUP | GRID | STEPS | SAVINGS |

**“Understand your solar before you buy your solar.”**

**English Edition** · Version 1.0 · October 2026

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# 01 About This Guide & What You Will Learn

| | |
|---|---|
| **Guide** | **SOS-01 — Solar Starter Guide: Understand Solar Before You Buy** |
| **Category · Level · Price** | Solar Fundamentals · Beginner · **Free** |
| **For** | First-time solar buyers — homes, farms, shops — and students |
| **Prerequisites** | Helpful: R1 Electricity Basics, R2 Your Bill, R3 Sun & Site |
| **Read next** | SOS-05 Homeowner Guide · SOS-12 Kisan Solar Guide · SOS-10 PM Surya Ghar · R5 Evaluating a Quotation |

### The problem

"Should I go solar? Which type? How much? What happens after I say yes?" Most first-time buyers start with a sales call — before they understand any of the answers.

### The knowledge gap

Buyers often don't know the difference between on-grid, off-grid and hybrid systems, what a subsidy does and doesn't cover, or what the steps from application to commissioning look like.

### The decision

After reading, you can decide whether solar suits you, which system type fits your needs, roughly what size, and what to expect from the process.

### The action

Complete the "Is solar right for me?" check, estimate your size, and prepare the document pack before contacting vendors.

**What you will learn:** system types · components · the Indian installation journey · subsidies at a glance · first size and savings estimates · how to value a quotation · warranties and red flags.

# 02 Why This Matters

Solar is a 20–25-year decision made in a few weeks. The right system cuts your bill for decades; the wrong one — wrong type, wrong size, poor installation — can disappoint for just as long. Understanding the basics first turns you from a person being sold to into a person choosing.

**WHAT CHANGES IN INDIA?**

Rooftop solar in India runs through your **DISCOM** (approval, net meter), often a **government scheme** (for homes, PM Surya Ghar: Muft Bijli Yojana), and an installer (vendor/EPC). Power cuts, slab tariffs, dust, heat and monsoon all shape what suits you.

# 03 Fundamentals — What a Solar System Is

| Component | What it does | What to look for |
|---|---|---|

| | | |
|---|---|---|
| **Solar panels (modules)** | Turn sunlight into DC electricity | Wp rating, make/model, certification, warranties |
| **Inverter** | Converts DC to AC for your home and the grid | kW AC rating, phase, efficiency, warranty |
| **Mounting structure** | Holds panels at the right angle for decades | Material, corrosion protection, wind design |
| **Balance of system** | Cables, DC/AC distribution boxes, fuses, breakers, surge protection | Listed sizes and ratings in the quote |
| **Earthing & lightning protection** | Safety for people and equipment | Number of earth pits, arrester, test report |
| **Net / bi-directional meter** | Counts units imported and exported | Installed or approved by your DISCOM |
| **Monitoring** | Shows generation in an app or display | Daily and total kWh visible to you |
| **Battery (optional)** | Stores energy for backup | Usable kWh, chemistry, warranty |

### Three system types

| | On-grid (grid-tied) | Hybrid | Off-grid |
|---|---|---|---|
| **Grid connection** | Yes | Yes | No |
| **Battery** | No | Yes | Yes (essential) |
| **Works in power cut?** | No — shuts down for safety | Yes, for selected loads | Yes — it is the only supply |
| **Cuts your bill?** | Yes | Yes | Replaces grid supply |
| **Relative cost** | Lowest | Higher (battery) | Highest per unit used |
| **Best for** | Reliable grid; goal is saving money | Frequent cuts and savings goal | No grid or very poor grid |

**SIMPLE**

On-grid saves money. Hybrid saves money and gives backup. Off-grid replaces the grid.

**DECISION**

Choose the type from your **main need**: savings, backup, or no grid. Paying for a battery you do not need is the most common way to overspend.

## 04 Science & Engineering — In Brief

**The photovoltaic effect.** A solar cell is a thin slice of specially treated silicon. Light knocks electrons loose; a built-in electric field inside the cell pushes them one way, creating DC current. Cells are wired into a panel; panels into strings.

**Why output varies.** Output follows sunlight strength, falls as cells heat up, and is reduced by dust, shade, wiring and conversion losses (R1, R3). Panel technologies — mono-crystalline PERC, TOPCon, bifacial and others — keep evolving. Compare **datasheets and warranties** rather than technology names.

**Why grid-tied systems dominate.** The grid acts like a giant battery: surplus daytime units are exported, evening units imported. No battery means lower cost and less maintenance — but no backup.

**SAFETY**

Panels are live whenever light falls on them; batteries store energy even when switched off. Installation, wiring, earthing and grid connection must be done by qualified, authorised professionals.

# 05 How It Works — The Indian Rooftop Journey




You lead steps 1–4. Steps 5–7 involve your DISCOM (and, for subsidised homes, the national portal). Step 8 is yours for 25 years. Exact steps, order and documents vary by state and scheme — follow your DISCOM's and the portal's current instructions.

**Figure 1.** The rooftop solar journey in eight steps, with the guide that helps at each step.

## Is solar right for me?

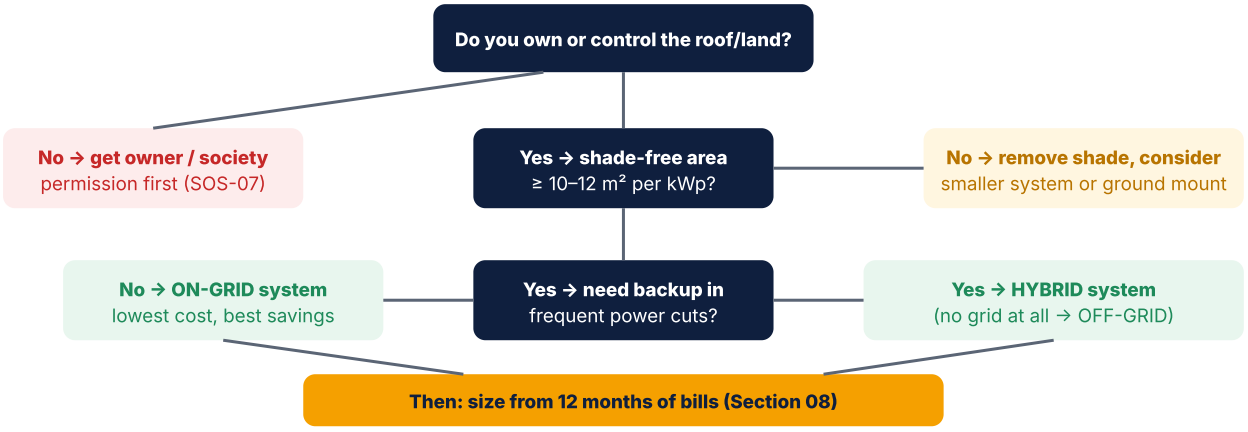



**Figure 2.** A first suitability check. A "No" is usually a problem to solve, not a dead end.

# 06 Indian Context — Schemes and Rules at a Glance

**VERIFY · PM SURYA GHAR: MUFT BIJLI YOJANA (RESIDENTIAL)**

| | |
|---|---|
| **What** | Central scheme for residential rooftop solar, launched February 2024, aiming at about 1 crore households |
| **Central financial assistance** | ₹30,000 per kW for the first 2 kW, ₹18,000 for the additional kW up to 3 kW — **maximum ₹78,000** for 3 kW and above |
| **Where** | National portal: **pmsuryaghar.gov.in** — registration, vendor selection, application tracking |
| **Conditions** | Eligibility, approved vendors and equipment requirements are set by the scheme — confirm the current conditions on the portal before you buy |

Checked 7 Oct 2026 against secondary sources (scheme summaries and news coverage of July 2026); the official portal could not be accessed from our research environment. Schemes change — confirm on the portal and with your DISCOM. Approval and payment are never guaranteed. Full step-by-step guide: SOS-10.

| Topic | What to know |
|---|---|
| **Farmers** | MNRE's PM-KUSUM programme has supported solar pumps and farm solar plants. Check its current status and your state's implementation (SOS-12, R9). |
| **State support** | Some states add their own incentives; others do not. Verify with your state nodal agency or DISCOM (SOS-11). |
| **Net metering** | Rules on capacity limits, settlement and surplus are set by your state regulator (SOS-08). |
| **Equipment quality** | Look for certified products; for subsidised systems, use equipment that meets the scheme's requirements. |
| **Space** | MNRE: about 10–12 m² of shadow-free area per kWp. |
| **Output** | Son of Surya regional planning figures: 2.9–4.1 units per kWp per day, depending on region (R1, R3). |

# 07 Practical Application — Prepare Your Pack

- [ ] 12 months of electricity bills (connection in your name)
- [ ] Property ownership proof or owner/society permission
- [ ] Rough roof sketch with south direction, obstacles and shade-free area (R3)
- [ ] Your needs: savings only, or backup too? Which loads, how many hours?
- [ ] Future plans: EV, more ACs, extra floor, pump

- [ ] Your first size estimate (Section 08) and budget range
- [ ] Identity and bank documents as required by the DISCOM or scheme

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**SON OF SURYA FREE TOOLS**

The **Son of Surya Solar Calculator** turns your bills and region into a first size and savings estimate. The free **Solar Project Passport** keeps your bills, quotes, approvals and warranties together — from first quote to 25 years of ownership.

# 08 Calculations

| C1 First size estimate | |
|---|---|
| FORMULA | `kWp ≈ yearly units ÷ 365 ÷ regional figure` |
| ASSUMPTIONS | Indore, Madhya Pradesh (figure 3.6) · 2,900 units/year from 12 bills |
| WORKING | 2,900 ÷ 365 = 7.95 units/day → ÷ 3.6 = **2.2 kWp** |

| C2 Central assistance under PM Surya Ghar (structure as published) | |
|---|---|
| FORMULA | `CFA = ₹30,000 × min(kW, 2) + ₹18,000 × min(max(kW − 2, 0), 1)` |
| WORKING | 2 kW → ₹60,000 · 2.5 kW → 60,000 + 9,000 = **₹69,000** · 3 kW or more → **₹78,000** |
| INTERPRETATION | Confirm current amounts and how fractional capacity is treated on the portal before relying on this. |

| C3 Yearly savings — and what a fair price looks like | |
|---|---|
| FORMULAS | `Savings/yr ≈ kWp × figure × 365 × ₹ per unit` · `Payback ≈ net cost ÷ savings/yr` |
| ASSUMPTIONS | **ILLUSTRATIVE EXAMPLE** 2.2 kWp, figure 3.6, all units used or credited at an assumed ₹8 per unit |
| WORKING | 2.2 × 3.6 × 365 = 2,891 units → × ₹8 = **₹23,128 per year**. Payback in 5 years would mean net cost ≈ 5 × 23,128 = ₹1,15,640; payback in 7 years ≈ ₹1,61,896. |
| INTERPRETATION | Use this **in reverse**: once you have real quotes, divide each net cost by your savings to compare paybacks. These are not price estimates. |

# 09 Indian Case Study

**ILLUSTRATIVE EXAMPLE — not a real customer or project**

**A family in Indore** uses 2,900 units a year and faces 1–2 hours of power cuts on some summer evenings. A neighbour recommended "a 5 kW system with batteries". Working through this guide, they found:

- **Size:** C1 suggests ≈2.2 kWp. Their 30 m² shade-free terrace holds ≈2.7 kWp — enough.
- **Type:** cuts are short and infrequent. Their existing home UPS already covers fans and lights. An **on-grid** system meets the savings goal; a hybrid would add battery cost for a need already met.
- **Subsidy:** at 2.2 kW, the published structure gives ₹60,000 + 0.2 × ₹18,000 = ₹63,600 — to be confirmed on the portal.
- **Process:** they prepared the document pack and chose to compare three registered vendors' written quotes.

> **OUTCOME**
>
> They requested quotes for 2.2–2.7 kWp on-grid systems with yearly kWh estimates and full component lists — and kept the hybrid option in mind if power cuts worsen.

# 10 Cost & Economics

| Cost drivers | Savings drivers |
|---|---|
| Panels and inverter (make, size) | Units generated (region, shade, tilt, cleaning) |
| Structure height and material | Your marginal tariff (R2) |
| Battery (hybrid / off-grid) | How exported units are credited (SOS-08) |
| Cables, protection, earthing | Subsidy received (if eligible) |
| DISCOM fees, meter, installation | Long system life with good maintenance |

**Lifecycle:** panels last decades with slow decline; inverters usually need replacing at least once during the system's life; batteries wear faster still. Include these in long-term thinking. **No saving or payback figure is guaranteed** — tariffs, rules and weather change.

# 11 Common Mistakes

| Mistake | Better approach |
|---|---|
| Starting with a sales call | Start with your bills and roof |
| Buying batteries "just in case" | Buy backup only for a real, defined need |
| Choosing on price per kW alone | Compare full scope, equipment and warranties |
| Assuming every vendor can process subsidies | Check the vendor is registered for the scheme |
| Oversizing for a "zero bill" | Size to yearly use; fixed charges remain |
| Ignoring maintenance | Plan cleaning and annual checks from day one |
| Not keeping documents | Keep every paper — warranties depend on them |

# 12 Buyer Protection

### A COMPLETE QUOTATION LISTS

- [ ] Panel make, model, Wp × quantity = kWp
- [ ] Inverter make, model, kW, phase
- [ ] Structure material and finish
- [ ] Cables, protection devices, earthing
- [ ] Yearly kWh estimate with assumptions
- [ ] Scope: who handles DISCOM/portal steps
- [ ] Payment stages and timeline

### WARRANTIES TO ASK FOR, IN WRITING

- [ ] Panel **product** warranty (defects)
- [ ] Panel **performance** warranty (output over years)
- [ ] Inverter warranty
- [ ] Installer **workmanship** warranty
- [ ] Who you call for service, and response time

### RED FLAGS

Guaranteed zero bills · guaranteed subsidy · no written estimate in kWh · large advance before approval with no written terms · pressure to sign today.

# 13 Practical Checklist

- [ ] I know my yearly units and daily average (R2)
- [ ] I know my shade-free area and roof condition (R3)
- [ ] I have chosen a system type from my real need: on-grid / hybrid / off-grid
- [ ] First size estimate: _____ kWp · region figure _____
- [ ] I have checked scheme eligibility and current conditions on the official portal
- [ ] I will compare at least three written quotes with the same scope
- [ ] I know which warranties I need in writing
- [ ] I have a folder (or Solar Project Passport) for every document

# 14 Problem Solving

| Level | Problem | Solution |
|---|---|---|
| EASY | A home needs savings only; grid is reliable. Which type? | **On-grid** — lowest cost for the goal. |
| INTERMEDIATE | Published CFA structure: what is the assistance for 1.5 kW? | 1.5 × ₹30,000 = **₹45,000** (confirm on the portal) |
| REAL-WORLD | Gujarat home: 3,600 units/year; 26 m² shade-free roof. Size by need, size by roof, and which limits? | Need: 3,600 ÷ 365 ÷ 3.8 = **2.6 kWp**. Roof: 26 ÷ 11 ≈ **2.4 kWp**. The **roof** limits; expect ≈ 2.4 × 3.8 × 365 ≈ 3,330 units/year — about 92% of use. |

# 15 FAQ

**Q1 How long does a system last?**
Panels are designed for decades; inverters and batteries need replacement sooner.

**Q2 Will solar work in monsoon?**
Yes, at lower output. Plan on yearly totals.

**Q3 Can I add panels later?**
Sometimes — depends on inverter capacity, roof and DISCOM approval.

**Q4 Do I need batteries?**
Only for backup or no-grid sites.

**Q5 Is a subsidy guaranteed?**
No. It depends on eligibility, scheme rules and approval.

**Q6 Who applies to the DISCOM?**
You or your vendor, as per scheme and DISCOM process — agree in writing who does what.

**Q7 Can tenants install solar?**
Only with the owner's permission and a connection that allows it.

**Q8 How often should panels be cleaned?**
It depends on dust; many sites need regular cleaning in dry months (SOS-20).

**Q9 Will solar damage my roof?**
Not if properly designed and installed with waterproofing care.

**Q10 Can I install it myself?**
No — grid-connected installation needs qualified, authorised professionals.

**Q11 What if my vendor closes down?**
Manufacturer warranties remain; keep all documents and serial numbers.

**Q12 Is solar worth it for small bills?**
Run C3 with your own numbers — very low use means longer payback.

# 16 Glossary

**BOS** — balance of system: everything except panels and inverter.

**CFA** — central financial assistance (subsidy).

**Commissioning** — final testing and start of operation.

**DISCOM** — electricity distribution company.

**EPC / vendor** — company that designs and installs.

**Hybrid system** — grid + battery + solar.

**Net meter** — records import and export.

**Off-grid** — no grid connection.

**On-grid** — connected to the grid, usually no battery.

**Payback** — years for savings to equal net cost.

**Performance warranty** — guaranteed output over years.

**Product warranty** — cover against defects.

**PM Surya Ghar** — central residential rooftop scheme.

**PM-KUSUM** — central farm solar programme.

**Workmanship warranty** — installer's cover on installation.

# 17 What To Do Next

SOS-01 ✓ → Free Calculator → SOS-05 Home / SOS-12 Kisan → R5 Quotes →

Solar Project Passport

# 18 Sources

| Source | Used for | Checked |
|---|---|---|
| PM Surya Ghar: Muft Bijli Yojana — pmsuryaghar.gov.in (official portal); scheme summary (Wikipedia, citing news sources); Business Today, 28 Jul 2026 | Launch, target, CFA structure, portal | 7 Oct 2026 (secondary) |
| MNRE Rooftop Solar FAQ via PIB (28 Apr 2022) | 10–12 m² per kWp; net metering | 7 Oct 2026 |
| Son of Surya regional planning figures (R1/R3 method) | Size and savings estimates | 7 Oct 2026 |
| MNRE — PM-KUSUM programme (mnre.gov.in) | Farmer scheme mention | status to verify |

# 19 Disclaimer

Published by Son of Surya for general education; not financial, legal or engineering advice. Scheme details are summarised from sources checked on the date shown and may have changed — always confirm on the official portal and with your DISCOM. Subsidy approval and payment are not guaranteed. Calculations are estimates on stated assumptions; examples marked ILLUSTRATIVE EXAMPLE are invented and are not price quotations. Installation and electrical work must be carried out by qualified, authorised professionals.

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# India’s structured solar knowledge library

Free foundation guides, practical calculators and honest, step-by-step knowledge — so every Indian home, farm and business can understand solar before buying it.

UNDERSTAND → CALCULATE → VERIFY → COMPARE → DECIDE → ACT

FREE TOOL

### Son of Surya Solar Calculator

Enter your bills and location. See system size, yearly units and savings — with every assumption shown.

FREE TOOL

### Solar Project Passport

Keep your solar project’s key numbers, documents and warranties in one place. An engineer-verified version is available.

**FREE FOUNDATION GUIDES**

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

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- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
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